

When Corruption is the Norm: Can Signaling Environmental Compliance Reduce Bribery?

Moogdho Mahzab^{1*}, Ayesha Ahmed², Nina Brooks³, Aprajit Mahajan⁴, Stephen Luby⁵, Debashish Biswas⁶,
Rofi Ahmed⁶, and Shoeb Ahmed⁷ [†]

September 23, 2026

Abstract

Firms in developing countries often operate in structurally informal environments where illicit payments are embedded in the cost of doing business. Using primary data from the brick kiln industry in Bangladesh, this paper studies whether credible signaling of regulatory compliance, beyond compliance itself, can reduce illicit payments. We exploit variation in banner display induced by randomized assignment to a broader environmental technology intervention to identify the causal effect of compliance signaling. Using variation in environmental intervention as an instrument, we estimate the local average treatment effect of banner display on illicit payments. While standard models predict that compliance signaling could increase rent extraction if officials perceive compliant kilns as more profitable, or reduce it if officials value regulatory compliance more, the findings support the latter. Each additional week of banner display reduces bribe payments and compliance costs per 1,000 bricks by about 1.0% and 0.3%, respectively, implying reductions of 18.6% and 5.4% over the average banner display period. In the binary specification, banner receipt reduces bribe payments by 19.3% and compliance costs by 5.2% over a firing season. The results suggest that credible compliance signals alter firm-bureaucrat interactions by limiting discretionary rent extraction in weak institutional settings.

Keywords: Brick Kilns, Signaling, Environmental Compliance, Corruption, Rent Extraction.

*Email: m.mahzab@cgiar.org.

^{†1}International Food Policy Research Institute; ²University of California, Davis; ³University of Michigan; ⁴University of California, Berkeley; ⁵Stanford University; ⁶International Centre for Diarrhoeal Disease Research, Bangladesh (icddr,b); ⁷Bangladesh University of Engineering and Technology.

Acknowledgments: We would like to thank Grant Miller, Abhijit Banerjee, and Shyamal Chowdhury for their valuable comments and suggestions on this research. We are grateful to Parastou Ghazi and Jyoti Das for their excellent research assistance. We also appreciate the helpful feedback we received from participants at seminars at Stanford University, the Pacific Conference for Development Economics at the University of California, Davis, and the Dhaka Winter Conference in Economics at BRAC Institute of Governance and Development (BIGD), BRAC University.

1 Introduction

Across much of the developing world, weak regulatory frameworks perpetuate a cycle of corruption and inefficiency in informal industries. To circumvent regulations, firms often rely on illicit payments to continue operating with outdated or unauthorized methods that not only suppress productivity but also harm the environment (Olken and Pande, 2012; Rodríguez-Pose and Di Cataldo, 2015). While a growing literature examines how anti-corruption strategies such as financial incentives, audits, and monitoring affect firm performance (Mattsson, 2025; Ha et al., 2024; Colonnelli and Prem, 2022), their effectiveness is frequently constrained by weak enforcement capacity and poor targeting of enforcement effort (Duflo et al., 2018). In this paper, we shift the focus to the firm side and ask whether firms can strategically reduce rent extraction by credibly signaling compliance. We argue that regulatory compliance alone may not suffice if it is not effectively communicated to bureaucrats through a credible signal.

We examine this question in the context of the brick kiln industry in Bangladesh, a sector marked by widespread regulatory violations, environmental degradation, and illicit transactions. Brick production in South Asia and parts of Sub-Saharan Africa remains largely informal and relies on traditional techniques that contribute to air pollution, poor health outcomes, and reduced agricultural productivity (Brooks et al., 2024; Miller et al., 2024; Kozlowski, 2021; World Bank, 2012). In Bangladesh, regulations governing kiln location, fuel use, and production technology are formally strict, yet violations are common. Our survey data show that 95 percent of kilns are located within one kilometer of agricultural land and more than 70 percent are situated near forests or gardens, in clear violation of siting rules. Interviews with kiln owners indicate that even when firms attempt to comply, final clearance depends on the discretion of local officials. This discretionary environment embeds illicit payments into the routine cost of operating a kiln.

To identify the causal effect of compliance signaling on illicit payments, we leverage a parallel intervention implemented alongside a large-scale randomized controlled trial in southwestern Bangladesh (Brooks et al., 2025). The broader RCT provided technical training and assistance to promote energy-efficient production practices. Within this framework, banners were distributed to kilns that verified adoption of key operational improvements. These banners served as credible and externally endorsed signals of regulatory compliance to public officials.

A central empirical challenge is that banner display and its duration are not randomly determined. Kilns that adopt cleaner practices and receive banners may differ systematically from others in ways that also affect illicit payments. To address this endogeneity, we implement an instrumental variables strategy. Specifically, we instrument banner exposure, measured either as duration of display in weeks or as a binary indicator for receipt, with randomized assignment to the environmental intervention.

Random assignment generates exogenous variation in the likelihood of banner eligibility and display, and remains orthogonal to unobserved determinants of illicit payments. This approach isolates variation in compliance signaling that is induced by experimental assignment rather than by kiln-level characteristics or relationships with officials. The resulting estimates can be interpreted as a local average treatment effect for kilns whose banner exposure responds to the randomized intervention.

Our conceptual framework draws on the economics of signaling. Following Spence (1973), a credible signal is an observable action that conveys otherwise hidden information about a firm's type to an uninformed party, though in our setting the signal is conferred through verified adoption and display rather than chosen strategically by the firm at a type-dependent cost. The predicted effect of such a signal on illicit payments is ambiguous. Bureaucrats may perceive compliant kilns as more productive and financially stronger, and therefore better able to pay higher bribes, in which case signaling compliance could increase rent extraction. Alternatively, officials may value the reduction in environmental harm or view compliant firms as less risky targets for enforcement, which could reduce bribe demands. Which force dominates is an empirical question, and is the central question this paper addresses.

Our empirical findings are consistent with the latter interpretation. We show that credible compliance signaling reduces illicit costs. Each additional week of banner display lowers bribe payments by roughly 1.0 percent and compliance costs by 0.3 percent per 1,000 bricks, implying reductions of 18.6 percent and 5.4 percent over the average 18.7-week display period. In the binary specification, banner receipt reduces bribe payments by 19.3 percent and compliance costs by 5.2 percent over a firing season. Political donations remain statistically unchanged across both specifications. These results indicate that compliance alone is not sufficient. Rather, it is the sustained and credible communication of compliance that reshapes interactions between firms and bureaucrats.

This study contributes to several strands of the literature on corruption and informal industries. We extend the literature on the industrial organization of corruption, which studies how illicit payment demands are set in interactions between firms and regulators (Svensson, 2003; Olken and Barron, 2009; Banerjee et al., 2012). We show that these illicit payments respond to a credible signal of compliance and are shaped by what officials can credibly infer about a firm. While Tamkoç (2024) argue that bribery greases the wheels of burdensome regulation, this interpretation fits our context poorly. Since nearly all kilns violate siting regulations, as discussed in Section 2, these payments are better understood as discretionary rents extracted by officials rather than as a price paid for regulatory relief. The closest prior evidence comes from Duflo et al. (2013), who show that third-party environmental auditing can reform auditors' incentives toward accurate reporting, make compliance information more credible, and reduce plant emissions. Their intervention reforms the incentives of a corrupt intermediary, the third-party auditor, so that the regulator receives truthful pollution reports. In our setting, a trusted third party verifies the firm's practices and issues a banner that signals compliance to officials. While

their analysis traces the effect through to plant emissions, we measure the effect on the illicit payments that officials extract.

We also add to the literature on measuring corruption in informal sectors, where the sensitivity and concealment of illicit payments make reliable data hard to obtain. Self-reports and perception measures can diverge substantially from underlying behavior, since officials tend to hide extraction where it is hardest to detect (Olken, 2009). In response, one strand of the literature develops objective measures that compare reported against actual quantities, such as “missing expenditures” in public works (Olken, 2007) and leakage of funds traced from government to frontline facilities (Reinikka and Svensson, 2004), while another shows that carefully designed micro-surveys can elicit quantitative corruption data directly from firms and service providers (Reinikka and Svensson, 2006). We contribute to this second, survey-based strand with detailed and disaggregated data on bribes, compliance costs, and political donations from over three hundred informal firms. The disaggregation lets us document how the intervention affects not only the level of illicit payments but their composition across these categories.

The study also speaks to the design of anti-corruption strategy in weak-enforcement settings (Islam, 2016). Where command-and-control regulation is poorly enforced, developing countries often rely on informational and voluntary instruments such as public disclosure of firms’ environmental performance (Blackman, 2010). Evidence that such instruments work is mixed, and a recurring concern is that they may depend on strong background regulation to be effective (Blackman, 2010). Our banner is an informational signal of this kind, and we show it reduces the illicit payments officials extract even where formal enforcement is weak. This complements top-down monitoring, which can be costly and whose productivity effects may be regime-dependent or even adverse when corruption is entrenched and bureaucratic delay is the binding margin (Ha et al., 2024). We show that a bottom-up, firm-side compliance signal can align firm incentives with regulatory objectives through a low-cost mechanism, offering a complement to enforcement-based approaches in settings where state capacity is weak.

The paper is organized as follows. Section 2 outlines the regulatory framework for kiln siting in Bangladesh. Section 3 presents the conceptual framework. Section 4 describes the study design, data sources, and balance tests. Section 5 details the empirical strategy. Section 6 presents the main IV results and robustness checks. Section 7 concludes.

2 Regulatory Environment and Non-Compliance

This section outlines the statutory siting requirements for brick kilns in Bangladesh. We then present descriptive evidence on compliance status among our sample and demolished kilns, and draw on interviews with officials to examine how discretionary enforcement facilitates an environment where

illegal operations can persist.

2.1 Brick Manufacturing and Kiln Establishment Act

The 2013 Brick Manufacturing and Kiln Establishment Act (amended in 2019) mandates compliance with environmental siting regulations and prohibits the Department of Environment, or any other competent authority, from issuing permits, licenses, or environmental clearance certificates to brick kilns located in restricted zones (Figure A.1). Under the Act, kilns may not be established within one kilometer of designated “prohibited areas,” including residential and commercial areas, agricultural land, private forests or gardens, ecologically critical areas, or within two kilometers of the boundary of a public forest. Kilns are also prohibited on hilltops, slopes, or adjacent hill surfaces unless located at least 0.5 kilometers from the base. In addition, the Act requires a minimum distance of one kilometer between brick kilns and specified public infrastructure, including railways, educational institutions, hospitals, research centers, and related facilities.

The Act further provides statutory definitions for key environmental and land use categories relevant to siting restrictions. A “residential area” is defined as any locality comprising at least 50 households. “waterbody” refers to land that remains submerged for six months or more during the year. A “garden” includes areas containing at least 100 fruit or forest trees per hectare and explicitly encompasses tea gardens. “Privately owned forest” denotes forestland formally recognized by the Forest Department with canopy cover exceeding 30 percent, including social and village forests, while a “public forest” is defined residually as any forest not classified as privately owned.

2.2 Regulatory Violations in Kiln Siting

Although the 2019 amendment to the Brick Manufacturing and Kiln Establishment Act granted a two-year grace period for the relocation of non-compliant kilns, our 2023 data indicate that widespread violations persist. As shown in Figure 2.1, the average kiln is located 151 meters from agricultural land, 384 meters from residential areas, and 629 meters from forests, below the legally mandated one kilometer thresholds. Kilns that were demolished during the survey period do not differ systematically on most dimensions, though they are, on average, located closer to schools and waterbodies.

Figure 2.1: Average distances from kilns to regulated areas.

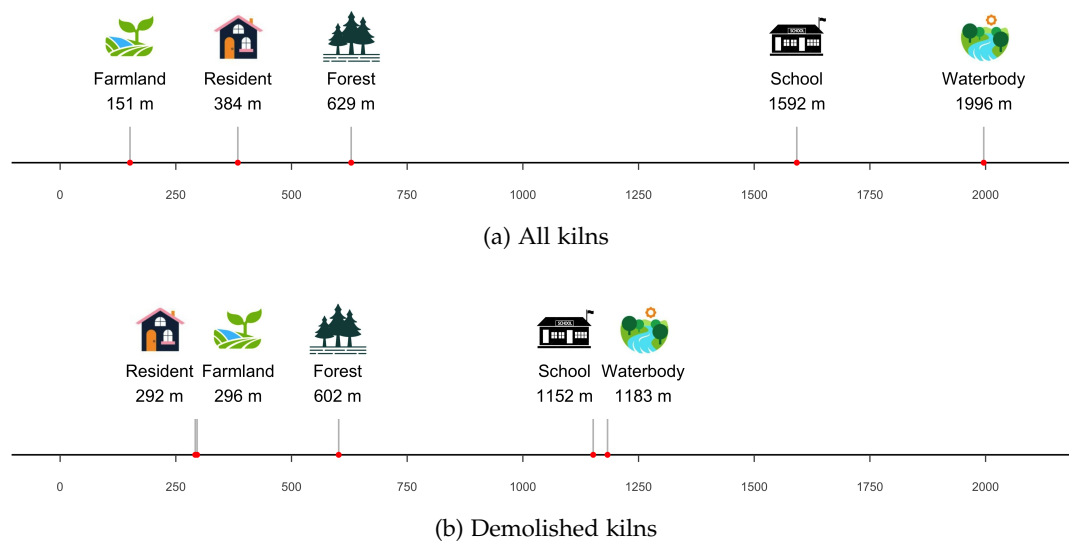


Table 2.1 reports the prevalence of siting violations and related compliance indicators for all kilns and for the subset that were demolished. Despite the mandated restrictions, 95% of kilns are sited within one kilometer of agricultural land, 85% within one kilometer of residential areas, and over 70% within restricted proximity to forests or gardens. Other regulatory provisions also show high rates of non-compliance, with 54% of kilns failing to meet the vegetation coverage requirement and 99% lacking a waste-management system. Demolished kilns exhibit similar patterns but with higher prevalence in several categories, including vegetation cover, where 87% fall below the required threshold. Of the 326 kilns with recorded demolition status, only 23 (7%) were demolished during the season, suggesting that enforcement remains limited even two years after the expiration of the grace period set by the 2019 amendment.

Table 2.1: Prevalence of statutory siting and environmental violations

Violation type	All kilns		Demolished Kilns	
	<i>n</i>	%	<i>n</i>	%
<i>Siting violations</i>				
Within 1 km of a school	50	15.2	8	34.8
Within 1 km of a waterbody	110	33.5	12	52.2
Within 1 km of a forest or garden	231	71.1	17	73.9
Within 1 km of a residential area	278	85.3	20	87.0
Within 1 km of agricultural land	305	94.7	19	82.6
<i>Non-siting violations</i>				
Tree or vegetation cover below 30%	178	54.3	20	87.0
Lacks waste-management system	325	99.1	23	100.0

Notes: The sample contains 357 kilns. *n* is the number of kilns in violation; percentages are computed using non-missing observations for each row. Item-specific missing counts in the full sample are: school, 29; waterbody, 29; forest or garden, 32; residential area, 31; agricultural land, 35; tree or vegetation cover, 29; and waste-management system, 29. The *Demolished kilns* columns report violation prevalence among the 23 demolished kilns.

Table 2.2 presents the distribution of kilns by the number of siting regulations violated, which ranges from zero to five. The cumulative columns report the share of kilns committing at least the stated number of violations. Nearly all kilns, 99.7% of the sample, breach at least one siting restriction, and about three-quarters (74.7%) are in breach of three or more. Only a single kiln is fully compliant. Kilns violate 3.0 siting regulations on average, rising to 3.3 among those that were demolished, where violations are concentrated at the upper end of the distribution.

Table 2.2: Distribution of the number of siting violations per kiln

No. of violations	All kilns (<i>N</i> = 328)		Demolished kilns (<i>N</i> = 23)	
	%	Cumulative %	%	Cumulative %
5	1.8	1.8	4.3	4.3
4	25.6	27.4	47.8	52.2
3	47.3	74.7	26.1	78.3
2	18.6	93.3	17.4	95.7
1	6.4	99.7	4.3	100.0
0	0.3	100.0	—	—
Total	100.0		100.0	

These findings highlight the widespread nature of statutory non-compliance in the brick industry, and the limited enforceability in regulating kiln siting and operations. Although a small subset of kilns were demolished, it remains unclear whether these were driven by legal violations per se or by the failure of kilns to make informal payments needed to continue operation under non-compliance. More broadly, the persistence of violations suggests that existing regulations may be poorly aligned with operational feasibility, along with population density and demand for bricks within the country.

2.3 Insights from the Interviews with Government Officials

This section draws on 24 qualitative interviews with public officials conducted across six districts in Khulna Division, Bangladesh, where the intervention took place. The respondents included representatives from several key government departments: the Department of Environment (DoE, 9 officials), Upazila Nirbahi Offices (UNO, 8 officials), Deputy Commissioner Offices (DC, 5 officials), the Water Development Board (1 official), and the Labor Department (1 official), as shown in Table A.1. These interviews were designed to capture officials' perceptions and experiences related to brick kiln operations and environmental compliance. The interviews helped us to understand the entire ecosystem of public officials' engagement with the brick kilns. We did not set a specific target for the number of qualitative interviews; instead, we reached out to all the relevant government agencies responsible for brick kiln operations and were able to complete a total of 24 interviews. This level of participation is noteworthy given that public officials are generally reluctant to discuss brick kiln operations, as it is an open secret that the sector has long been a cash-cow for easy targeting and extracting bribes.

Across departments, officials consistently cited air pollution as the most serious and visible problem associated with brick kilns, especially the discharge of black smoke during the brick-making season. This concern is emphasized as the primary trigger for regulatory actions, such as inspections, fines, and, in some cases, closure of non-compliant kilns. Other frequently cited concerns include the use of fixed and drum chimney kilns, topsoil extraction from agricultural land, operation without environmental clearance, reliance on firewood, and kiln siting in close proximity to protected areas.¹ Government officials described that actions taken against kilns are typically motivated by a combination of these factors, with specific cases sometimes driven by citizen complaints, legal non-compliance, or proximity to sensitive sites.

The DoE officials mentioned that they have stopped operations of 13 kilns this season—11 of them were using fixed chimney technology, and 2 used drum chimneys. It is important to note that both drum chimney kilns, which were banned in 2002, and fixed chimney kilns, whose ban process began in 2012 and was implemented gradually over time, are among the most polluting kiln technologies. Despite these official bans, such kilns are still observed operating in many areas. In our intervention sample, we selectively chose only kilns using the zigzag method and coal as the main fuel for burning green bricks.

Officials from the DoE noted that their main responsibilities include routine and unplanned inspections, responding to complaints, issuing notices, recommending penalties, and coordinating with the DC office and law enforcement. UNO and DC offices often collaborate in joint operations and mobile courts to enforce regulations, particularly against illegal or hazardous kilns. However, limitations

¹Fixed and drum chimney kilns are older kiln technologies associated with higher particulate emissions relative to zigzag kilns. Topsoil extraction refers to the removal of fertile surface soil from agricultural land for brick production. Firewood use contributes to local deforestation.

such as insufficient resources, local political interference, and logistical barriers can hinder sustained monitoring and enforcement.

In summary, our interviews with 24 public officials suggest that while air pollution remains the leading trigger for regulatory concern and enforcement activity, kiln oversight is shaped by a broader set of considerations related to environmental protection, legal compliance, and community well-being. Across interviews, it became evident that enforcement is influenced by a combination of technical criteria, field-level realities, political dynamics, and bureaucratic discretion. Weak monitoring systems, inconsistent implementation of regulations, and the discretionary authority of frontline officials create conditions under which rent-seeking and bribery can persist. The interviews portray the brick kiln sector as a setting where corruption becomes embedded within everyday regulatory interactions, particularly in an environment characterized by weak institutional capacity and limited accountability. This pattern reflects a broader regularity in developing economies, where heavy regulation of entry and operation is associated with higher corruption rather than better outcomes, consistent with the view that permits function in part as instruments for extracting rents (Amin and Soh, 2021; Djankov et al., 2002).

All the public officials we interviewed mentioned that they were aware of our intervention. They said they had noticed banners in front of some kilns, which signaled that these kilns were adopting our approach—referred to at the time as the “digital method.” The officials recognized that this method could help reduce pollution and lower overall coal consumption. They also mentioned that the potential for higher profits from adopting this method, by reducing coal use, was a good motivation for them to adopt the interventions. We draw on these insights from the interviews with public officials and use them in the next section to develop a simple signaling model as a conceptual framework.

3 Conceptual Framework-Signaling Model: Officer Preference, Extra Profit, and Banner Effect

The banner is an observable signal that conveys otherwise hidden information about kiln compliance to officials. The conceptual framework combines two literatures. The officer’s extraction decision follows the canonical model of corruption as monopolistic rent-setting under a convex cost of extraction (Shleifer and Vishny, 1993; Svensson, 2003), in which the bribe demanded rises with the firm’s perceived ability to pay and falls with its resistance. The banner’s role follows the economics of certification, in which a credible third-party intermediary transforms an unobservable firm type into an observable signal (Lizzeri, 1999; Dranove and Jin, 2010). Unlike the firm-chosen signal of Spence (1973), the banner is conferred through external verification. The signal enters a bureaucrat’s extraction problem directly, so that a change in what the officer infers about the firm shifts the rent demanded.

Model Setup

We consider a kiln that may display a banner indicating verified adoption of Zigzag 2.0. Let E_i denote banner display. In the binary case, $E_i \in \{0, 1\}$ indicates whether kiln i displays the banner; in the continuous case, $E_i \geq 0$ measures display duration in weeks. Because banners are assigned through the randomized intervention, E_i is exogenous to the kiln's unobserved willingness to signal compliance. The officer observes E_i and demands an illicit payment $b_i \geq 0$.

Banner display affects bribe extraction through two opposing channels. First, adoption may increase kiln profitability by $\Delta > 0$. If officers perceive adopting kilns as having higher surplus, banner display may increase the marginal return to extraction. This is captured by $\lambda \Delta E_i$, where $\lambda \geq 0$ measures the extent to which perceived additional surplus is translated into higher bribe demands. Second, the banner communicates compliance with an environmentally beneficial technology. Officers with stronger environmental orientation may be less willing to extract rents from visibly compliant kilns. Let $\beta \in [0, 1]$ denote the officer's environmental orientation, and let $\gamma > 0$ denote the value attached to the pollution reduction associated with adoption.

The officer chooses b_i to maximize,

$$U_O(b_i, E_i) = (1 - \beta + \lambda \Delta E_i - \beta \gamma E_i) b_i - \frac{1}{2} \kappa b_i^2,$$

where $\kappa > 0$ captures the marginal cost of extracting larger bribes, reflecting punishment risk, bargaining frictions, reputational costs, or firm resistance. The expression in parentheses is the net marginal return to extraction, where $(1 - \beta)$ is the return in the absence of the signal. Banner display may raise this return through perceived additional surplus, $\lambda \Delta E_i$, or lower it by making environmental compliance salient, $\beta \gamma E_i$. The quadratic term imposes convex costs of extraction and yields an interior optimum.

Optimal Bribe and Comparative Statics

The officer's first-order condition is

$$\frac{\partial U_O}{\partial b_i} = 1 - \beta + \lambda \Delta E_i - \beta \gamma E_i - \kappa b_i = 0,$$

which yields the optimal bribe demand

$$b_i^*(E_i) = \frac{1 - \beta + \lambda \Delta E_i - \beta \gamma E_i}{\kappa}.$$

Banner Display

In the binary case, the optimal bribe is $b_i^*(0) = (1 - \beta)/\kappa$ without banner display and $b_i^*(1) = (1 - \beta + \lambda\Delta - \beta\gamma)/\kappa$ with display. The effect of banner display on bribe extraction is therefore

$$b_i^*(1) - b_i^*(0) = \frac{\lambda\Delta - \beta\gamma}{\kappa},$$

so banner display reduces bribe extraction when

$$\beta\gamma > \lambda\Delta.$$

This condition has a simple interpretation. The banner may increase bribe demands if officers interpret adoption as evidence of greater profitability and therefore greater extractable surplus, a force captured by $\lambda\Delta$. The banner may instead reduce bribe demands by credibly communicating environmental compliance, a force captured by $\beta\gamma$ that is stronger when officers place greater value on environmental improvement. The net effect of banner display is negative when the compliance-signaling channel dominates the perceived-surplus channel.

The model generates an ambiguous theoretical prediction. For officers with low environmental orientation, the perceived-surplus channel may dominate, and banner display increases bribe demands. For officers with sufficiently high environmental orientation, the compliance-signaling channel dominates, and banner display reduces bribe demands. The differential response across officer types is summarized by

$$\frac{\partial (b_i^*(1) - b_i^*(0))}{\partial \beta} = -\frac{\gamma}{\kappa} < 0,$$

so the bribe-reducing effect of banner display is stronger where officers are more environmentally oriented. We note that β affects extraction both through the baseline term $1 - \beta$ and through the interaction $\beta\gamma E_i$, so environmentally oriented officers extract less even absent a banner and also respond more strongly to one. Empirical strategies that exploit variation in officer orientation therefore capture a combination of level and slope differences.

Continuous Banner Display

We next allow banner display to be continuous, which connects the framework to our empirical measure of display duration. Let the compliance-signaling channel operate through $\beta\gamma g(E_i)$, where $g(\cdot)$ is increasing and concave with $g(0) = 0$. Concavity captures the idea that the credibility of the signal accumulates as the banner persists, but with diminishing increments. The optimal bribe becomes

$$b_i^*(E_i) = \frac{1 - \beta + \lambda\Delta E_i - \beta\gamma g(E_i)}{\kappa},$$

and the marginal effect of an additional week of display is

$$\frac{\partial b_i^*}{\partial E_i} = \frac{\lambda\Delta - \beta\gamma g'(E_i)}{\kappa}.$$

Because $g'(E_i)$ is positive and declining, the cumulative compliance signal increases with display duration, but the incremental bribe-reducing effect of an additional week diminishes over time. The linear specification used above is the special case $g(E_i) = E_i$. In the empirical analysis, we estimate the average per-week effect of banner display duration on illicit payments and examine whether the observed pattern is more consistent with the compliance-signaling channel or the perceived-surplus channel.

4 Data and Descriptive Statistics

This section describes the study context, data sources, and sample construction. It then presents descriptive statistics on illicit payments, discusses potential measurement concerns, and reports balance tests and sectoral scale.

The banner signaled adoption of a set of operational improvements to zigzag kilns, developed and evaluated in a parallel randomized trial (Brooks et al., 2025). The improvements required no change in kiln structure or capital equipment and instead modified firing practice, including less dense brick stacking with multiple zigzag air paths, continuous coal feeding by a single fireman, thicker ash insulation, and the use of biomass in newly inducted chambers. By improving combustion and reducing heat loss, these practices lower both fuel use and emissions. Among kilns that adopted them, the intervention reduced carbon dioxide emissions and fine particulate (PM_{2.5}) emissions by roughly 20 percent and specific energy consumption by about 24 percent, while raising the share of high-quality bricks and lowering fuel costs (Brooks et al., 2025). Total annual brick production and the number of firing circuits were unaffected, so the intervention changed how bricks were fired rather than how many were produced (Brooks et al., 2025). Because the improvements cut fuel spending, adoption was profitable for owners, which is what made the compliance signal credible to officials who understood that adopting kilns was both cleaner and more efficient.

4.1 Study Timeline and Data Collection

During the brick firing season from November 2022 to May 2023, a parallel randomized intervention was implemented to improve energy efficiency and reduce emissions in brick kilns (Brooks et al., 2025). The study covers 357 kilns across six districts in Bangladesh (Jessore, Khulna, Jhenaidah, Chuadanga, Kushtia, and Narail), which were randomly assigned to a control group or one of two treatment arms, stratified by district and baseline productivity (Figure A.2). Kilns assigned to either treatment arm received information, technical support, and training to adopt several efficiency-enhancing practices,

including operational improvements related to coal feeding and brick stacking inside the kiln. Kilns in the ‘technical + incentive’ arm additionally received information emphasizing the business rationale for incentivizing workers to adhere to these practices.

Among treated kilns, we verify eligibility for receiving banners between November 2022 and February 2023 and distribute banners conditional on adoption of (1) proper brick stacking to improve airflow and combustion, and (2) use of a single fireman to manage kiln firing. We recognize 132 kilns² with banners indicating participation in a research initiative endorsed by nationally reputed institutions (Figures 4.1 and A.3). Importantly, the names displayed on the banners served as a credible signal of legitimacy and compliance. Institutions such as icddr,b and BUET are widely recognized in Bangladesh for their longstanding contributions to public health, engineering, and scientific research. During field visits, we observed that even in relatively remote rural areas, local communities and public officials expressed a high degree of trust and respect toward institutions associated with the intervention.

The timing of banner distribution generated substantial cross-kiln variation in banner display duration through the end of the firing season, ranging from 3 to 34 weeks. Although some control kilns independently adopted these practices, banners were distributed only to kilns assigned to the intervention arms (Table A.4). Variation in banner exposure arose because adoption of the recommended practices occurred at different points in the production cycle across kilns. Brick kilns do not begin operations simultaneously; rather, the start of production depends on kiln-specific factors such as the arrival of seasonal labor, access to coal and other key inputs, and readiness to initiate brick firing. As a result, kilns became eligible for verification and banner distribution at different times during the season, generating plausibly exogenous variation in the duration of visible signaling.

²Banner eligibility was based on enumerator observation of adoption using a structured checklist administered between November 2022 and February 2023 during the ongoing brick firing season. Continued practice was assessed using the same checklist during the kiln performance assessment (March–May 2023), with 108 out of 132 banner-recipient kilns continuing to practice the recommended methods.

Figure 4.1: The banner distributed to kilns that adopted environmentally friendly practices.



Note: The banner shown is an English translation of the Bangla language banner displayed in the kilns. It includes the logos of nationally reputed institutions, such as icddr,b and BUET, to credibly signal environmental compliance.

We draw primarily on baseline and endline surveys conducted as part of the broader RCT. These surveys provide the key outcome measures, including bribe payments, compliance costs, and political donations. All outcomes were self-reported by kiln owners and standardized per 1,000 bricks to ensure comparability across kilns with different production scales. Additionally, we administered a supplemental survey to collect kiln characteristics and proximity measures. To understand regulatory practices and enforcement dynamics in the sector, we also conducted qualitative follow-up interviews with government officials after the firing season. The study timeline, including the sequencing of surveys and intervention rollout, is shown in Figure A.4.

Figure A.5 summarizes the sample construction. Baseline data were collected from 357 kilns randomly assigned to one of three groups, control ($n=117$), technical assistance ($n=119$), and technical + incentive ($n=121$). Of these, 21 kilns were excluded from the endline sample due to data inconsistencies, resulting in a final sample of 336 kilns. Although 23 kilns were demolished during the season, they are retained in the endline sample because demolition occurred post-season and endline data were collected before closure. We further exclude 21 observations with missing kiln characteristics, resulting in a final analysis sample of 315 kilns, with 105 kilns in each intervention arm. Of the 315 kilns in the final analysis sample, 123 received banners, with 62 in the technical arm and 61 in the technical + incentive arm.

Table 4.1: Distribution of Analysis Sample by Environmental Intervention and Banner Status

Banner Status	Environmental Intervention Arm			Total
	Control	Technical	Incentive	
No Banner	105	43	44	192
Banner	0	62	61	123
Total	105	105	105	315

Notes: Banners were distributed only to treated kilns that met adoption criteria.

4.2 Illicit Cost Summary

In this study, continuous field presence and relationship-building with kiln operators, along with support from local kiln associations, reduced reluctance to disclose sensitive information. As a result, kiln owners were generally willing to discuss the informal payments required to operate throughout a production season. The field team from icddr,b, which collected the sensitive information, had developed an exceptional level of trust and rapport with brick kiln owners and brick kiln associations across Bangladesh. Researchers and technical experts from Stanford University and icddr,b had been engaged with the brick kiln sector in Bangladesh for nearly a decade through research, technical support, and environmental improvement initiatives. This long-term engagement helped establish credibility and familiarity between the research team and kiln stakeholders. Many kiln owners and managers viewed the research team as providing technical guidance intended to improve kiln production practices rather than as actors connected to regulatory enforcement. Consequently, respondents were less hesitant to discuss issues related to bribery, informal payments, and corruption. They were also reassured that the information collected would remain confidential and would be used strictly for academic research purposes rather than for formal investigations or punitive actions by authorities.

The survey included a module on illicit costs that asked kiln owners whether they made unofficial payments to public officials or local authorities and, if so, the amounts paid. Approximately 77% of kilns reported paying bribes, while nearly all reported compliance costs and political donations. Compliance costs include formal fines and fees associated with obtaining certificates or permits, though many owners were uncertain whether these payments were fully remitted to the government or partially diverted by officials. Political donations were made to local party leaders who exert influence over kiln operations, and in many cases kiln owners themselves held positions within local party structures, which blurs the boundary between political payments and regulatory compliance. The figures reported in Table 4.2 likely understate the true extent of illicit payments due to respondent hesitation and ambiguity over whether some payments constituted official compliance fees or payments to permit illegal operations. For this reason, we categorize bribes and compliance costs as official illicit payments and political donations as unofficial illicit payments.

Table 4.2: Proportion of Kilns Making Illicit Payments

Payments Type	Total Kiln	N (%)
Bribes	336	258 (76.8)
Compliance Cost	336	336 (100.0)
Political Donations	336	316 (94.0)

Notes: The sample comprises the 336 kilns reporting illicit-cost data at both baseline and endline. The instrumental-variables analysis uses the 315 of these kilns with complete data on all model covariates; reported payment rates are stable across the two samples.

Table 4.3 reports average illicit costs at two survey periods, measured both per production season and per 1,000 bricks. Total illicit costs declined over time—from 1,009,782 BDT per season at baseline to 762,465 BDT at endline, driven by reductions in all three categories. Despite these declines, illicit payments remained a nontrivial share of production costs.

Table 4.3: Illicit Costs at Baseline and Endline (N = 336)

Cost Type	Baseline (2022)		Endline (2023)	
	Per Season	Per 1,000 Bricks	Per Season	Per 1,000 Bricks
Bribes	202,391	46	144,996	27
Compliance Cost	658,576	155	546,811	101
Political Donations	148,815	32	70,658	13
Illicit Share of Total Costs [†]	3%	3.2%	1.9%	2%

Notes: The sample comprises the 336 kilns reporting illicit-cost data at both baseline and endline; the instrumental-variables analysis uses the 315 with complete covariate data, and estimates are stable across the two samples. [†]The illicit share of total costs is the sum of the three payment types divided by total kiln operating costs measured in the broader study.

Measurement of Illicit Costs

One potential concern in the collection of illicit cost data is the possibility of Hawthorne-type reporting effects across treatment and control groups. Specifically, kiln owners in the technical and technical+incentive treatment groups may have responded differently from those in the control group because the former received training and support that could positively affect kiln production practices. In principle, this could generate differential reporting behavior, where either control-group kilns or intervention-group kilns systematically overstate or understate the true amount of informal payments and political donations incurred during the production season.

We believe such concerns are unlikely to meaningfully bias our estimates for several reasons. First, control-group kilns were informed that technical training and support would gradually be extended to them in future phases of engagement. As a result, we do not believe that feelings of exclusion or dissatisfaction would systematically induce control-group respondents to misreport illicit payments relative to treated kilns. Second, as discussed earlier, the field enumerators and trainers from icddr,b

had built long-standing and highly trusted relationships with brick kiln owners and managers through years of engagement with brick kiln associations across Bangladesh. This credibility substantially reduced respondents' hesitation in discussing sensitive matters related to bribery, informal payments, and political donations.

Moreover, if any misreporting did occur, we believe it is more likely to have taken the form of underreporting rather than overreporting. Kiln owners and managers may have perceived very large reported payments as potentially reflecting poorly on their own operations or exposing them to reputational concerns. Importantly, there is little reason to expect such underreporting to differ systematically across treatment and control groups. Under this scenario, any remaining measurement error would likely attenuate the estimated treatment effects toward zero rather than inflate them. Consequently, the IV estimates presented in the paper, if biased at all, are more likely to represent conservative lower-bound estimates of the true effects.

4.3 Association of illicit costs with kiln characteristics

To assess whether illicit payments are systematically targeted to kiln characteristics, we first regress three types of illicit payments on a set of observable kiln characteristics, controlling for district fixed effects (Table 4.4). The results show limited evidence of systematic targeting: only two variables, distance to forest or garden, and the share of land with plants and trees, are significant for bribe payments and compliance costs.

Table 4.4: Association Between Individual Kiln Characteristics and Illicit Payments (District FE)

	<i>Bribe Payments</i>	<i>Compliance Costs</i>	<i>Political Donations</i>
Educational Institute Distance	0.002 (0.002)	0.001 (0.005)	0.002 (0.002)
Waterbody Distance	-0.000 (0.002)	0.001 (0.005)	0.001 (0.001)
Forest/Garden Distance	-0.006** (0.003)	-0.012* (0.006)	0.005 (0.004)
Residential Area Distance	-0.005 (0.006)	-0.002 (0.012)	0.000 (0.005)
Farmland Distance	0.001 (0.002)	0.008 (0.008)	0.001 (0.001)
Plants and Trees (%)	0.752*** (0.274)	-0.621 (0.380)	0.146 (0.140)

Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

To further test for the underlying selection pattern, we construct an index using the principal component of the kiln characteristics and regress this measure on illicit payment outcomes (Table 4.5). Across all specifications, the association between the index of violations and illicit payments is statistically insignificant.

Table 4.5: Association Between Principal Component of Kiln Characteristics and Illicit Payments.

	<i>Bribe Payments</i>	<i>Compliance Costs</i>	<i>Political Donations</i>
Violation Index	-0.859 (1.913)	-1.249 (3.308)	2.425 (1.868)
Observations	316	316	316
R-squared	0.039	0.021	0.021
Fixed Effect	District	District	District
Standard Errors	Robust	Robust	Robust

Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

These findings together suggest that illicit payment targeting by officials is effectively random with respect to observable kiln characteristics. This pattern is understandable since any randomly selected kiln is almost certain to be in violation of the law (see Table 2.2). Officials face a positive probability of detecting infractions regardless of which kiln they target and do not need to concentrate on kilns with more violations; instead, they can extract bribes or impose compliance costs on virtually any kiln.

4.4 Balance test of baseline characteristics

Table A.2 presents the baseline characteristics by banner treatment status. Most observable characteristics, including distances from regulated areas, waste infrastructure, and illicit cost measures, are statistically similar across groups, suggesting successful balance. Notable exceptions include a greater average distance to waterbodies and higher vegetation cover among kilns that received banners. However, the joint F-test indicates no significant overall differences at baseline.

4.5 Sectoral Context and Scale

To interpret the magnitude of effects estimated per 1,000 bricks, it is useful to scale them to the level of a kiln and a season. Kilns in this setting are large operations. An average kiln in our sample produces roughly 6.2 million bricks over a single firing season (Brooks et al., 2025), firing them across about five production circuits between November and June. The per-1,000-brick estimates, therefore, correspond to substantially larger amounts at the kiln level. A reduction of one taka per 1,000 bricks scales to approximately 6,200 taka per kiln over the season, so the estimated effects on illicit payments, while modest per unit, accumulate into economically meaningful sums given the volume of production.

This scale also clarifies the stakes of the environmental externality. Bangladesh produces on the order of tens of billions of bricks annually, almost entirely in informal kilns, making the sector one of the country's largest sources of particulate and greenhouse-gas emissions (Brooks et al., 2024; World Bank, 2011). Interventions that change behavior at the per-brick margin thus aggregate into large absolute effects on both illicit costs and pollution.

5 Empirical Strategy

This section describes the instrumental variables strategy used to identify the causal effect of compliance signaling on illicit payments and discusses the assumptions required for a causal interpretation.

5.1 Estimation

Estimating the causal effect of compliance signaling on illicit payments raises an endogeneity concern because banners were distributed conditional on verified adoption of environmentally compliant practices. Banner display is therefore not randomly assigned, but reflects a firm’s decision and capacity to adopt new technology, which may correlate with unobserved determinants of illicit payments. To address this selection concern, we use a two-stage least squares (2SLS) strategy that instruments banner display with randomized assignment to the environmental intervention, Z_{is} . The first-stage and second-stage equations are,

$$B_{is} = \theta_0 + \theta_1 Z_{is} + \gamma_s + X'_{is} \Gamma + \varepsilon_{is}, \quad (1)$$

$$Y_{is} = \beta_0 + \beta_1 \hat{B}_{is} + \gamma_s + X'_{is} \Gamma + u_{is}. \quad (2)$$

Equation (1) represents the first stage, where the endogenous variable B_{is} denotes either the duration of banner display in weeks, capturing signaling intensity, or a binary indicator for banner receipt. The instrument Z_{is} indicates randomized assignment to the technical treatment arm, the technical plus incentive arm, or the control group. Equation (2) estimates the effect of predicted banner exposure on illicit payments per 1,000 bricks (Y_{is}). All specifications include subdistrict fixed effects (γ_s) to account for local enforcement variation, baseline illicit payment measures, and a vector of kiln-level covariates (X_{is}). These controls include proximity to schools, waterbodies, forests, residential areas, and agricultural land, as well as tree coverage and the presence of a septic tank waste management system. Standard errors are clustered at the level of environmental intervention randomization.

5.2 Identification Assumptions

Interpretation of β_1 as the Local Average Treatment Effect (LATE) relies on instrument relevance, independence, exclusion, and monotonicity. Relevance ($\theta_1 \neq 0$) follows from the experimental design that randomized assignment significantly predicts banner exposure, as shown in the first stage results in Tables 5.1 and 5.2. Assignment increases banner display duration by approximately 11 weeks and raises the probability of banner receipt by about 58 percentage points. Because assignment to the environmental intervention was randomized, Z_{is} is independent of potential outcomes.

The exclusion restriction requires that intervention assignment (Z_{is}) affects illicit payments only

through the banner signal. A potential concern is that the assignment also induces technology adoption, which is a prerequisite for receiving the banner. If adoption independently reduces illicit payments, for example, through observable improvements in emissions, then the exclusion restriction would be violated.

We argue that the direct effects of technology adoption on illicit payments are limited in this setting. Brick kilns operate under weak monitoring capacity, and local field officers lack the technical equipment needed to verify technology adoption for emissions reductions. In routine interactions with the officials, the banner serves as the salient and readily observable indicator of compliance status. Moreover, we control for observable compliance proxies, which consist of siting, vegetation cover, and waste management in all specifications. Consistent with this interpretation, we find that banner receipt reduces bribe payments even among kilns that did not maintain technological compliance at endline, suggesting that reductions in illicit payments are driven by signaling rather than clean technology adoption (more discussion in Section 6.2.2).

Given that banner eligibility required verification and provision by the implementing team, always-takers are mechanically ruled out. Kilns in the control arm were not eligible to receive banners, even if they adopted recommended practices independently. Assignment, therefore, weakly increases the probability of banner exposure, satisfying the monotonicity condition. Under these assumptions, β_1 identifies the causal effect of compliance signaling among kilns whose banner exposure is induced by randomized assignment.

Table 5.1: First-Stage Estimates of Environmental Treatment Assignment on Banner Display Duration.

	<i>Outcome: Banner Duration (Weeks)</i>		
	<i>Bribe Payments</i>	<i>Compliance Costs</i>	<i>Political Donations</i>
Env. Intervention = 1 (Technical)	11.318*** (0.337)	11.219*** (0.397)	11.257*** (0.484)
Env. Intervention = 2 (Incentive)	10.618*** (0.421)	10.817*** (0.501)	10.787*** (0.600)
Constant	4.321 (2.428)	8.024 (4.181)	4.212 (2.525)
Observations	315	315	315
R-squared	0.561	0.574	0.561

Notes: All models control for subdistrict fixed effects, kiln characteristics, and baseline outcome corresponding to each second stage outcome. Standard errors clustered at the environmental treatment unit level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 5.2: First-Stage Estimates of Environmental Treatment Assignment on Banner Receipt.

	<i>Outcome: Banner Display</i>		
	<i>Bribe Payments</i>	<i>Compliance Costs</i>	<i>Political Donations</i>
Env. Intervention = 1 (Technical)	0.584*** (0.018)	0.581*** (0.020)	0.585*** (0.021)
Env. Intervention = 2 (Incentive)	0.579*** (0.020)	0.585*** (0.020)	0.586*** (0.024)
Constant	0.369 (0.208)	0.456 (0.261)	0.353 (0.197)
Observations	315	315	315
R-squared	0.589	0.592	0.588

Notes: All models control for subdistrict fixed effects, kiln characteristics, and baseline outcome corresponding to each second-stage outcome. Standard errors clustered at the environmental treatment unit level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

6 Results

This section presents the main IV estimates followed by robustness checks and heterogeneity analysis.

6.1 Main Results

Table 6.1 reports 2SLS estimates in which banner display duration is instrumented by randomized assignment to the environmental intervention. The coefficient on banner duration is negative for both bribe payments and compliance costs. An additional week of display induced by assignment reduces season-level bribe payments by 0.29 taka and compliance costs by 0.31 taka per 1,000 bricks. Relative to the illicit costs in a full firing season, one additional week corresponds to a reduction of roughly 1.0% in bribe payments and 0.3% in compliance costs, so a full season of banner display, about 18.7 weeks for treated kilns, implies reductions of about 18.6% in bribes and 5.4% in compliance costs. The effect on political donations (Column 3) is not statistically significant.

These signs map onto the conceptual framework in Section 3. A negative coefficient on banner duration indicates that the compliance-signaling channel, $\beta\gamma g'(E_i)$, dominates the perceived-surplus channel, $\lambda\Delta$, at the margin, so that the net marginal effect $(\lambda\Delta - \beta\gamma g'(E_i))/\kappa$ is negative. The linear specification estimates the average per-week effect under the special case $g(E_i) = E_i$ and does not by itself identify the diminishing increments implied by a concave $g(\cdot)$; we therefore read the estimate as an average signaling effect over the observed range of display durations rather than as a test of concavity.

The framework also speaks to why the effect differs across the three types of payment. Bribes correspond most directly to the discretionary rent b_i extracted by the enforcing officer, and it is here

that the signaling effect is strongest. Compliance costs are formal fees and fines that are only partly discretionary, consistent with their weaker and less robust response. Political donations are transfers to local political actors who are not responsible for environmental enforcement and are less directly subject to the officer's discretion, consistent with the null effect. The pattern across the three categories therefore matches the model's prediction that the banner operates on the discretionary-rent margin rather than on transfers across all domains.

Table 6.1: 2SLS Estimates of Banner Display Duration on Illicit Payments per 1,000 Bricks.

	<i>Bribe Payments</i>	<i>Compliance Costs</i>	<i>Political Donations</i>
Banner Duration (Weeks)	-0.294** (0.126)	-0.312** (0.157)	-0.051 (0.054)
Constant	39.847*** (5.059)	141.949*** (13.680)	12.798*** (2.029)
Observations	315	315	315
R-squared	0.193	0.235	0.211
Control mean (week) [†]	1.58	5.76	0.69

Notes: Two-stage least squares estimates. Banner display is instrumented by randomized assignment to the environmental intervention. All specifications include subdistrict fixed effects, kiln characteristics, and baseline illicit costs. † Control mean (week) is the control-group mean divided by the average banner duration among treated kilns (18.7 weeks). Standard errors clustered at the environmental-intervention randomization unit are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 6.2 reports 2SLS estimates in which banner display, a binary indicator for whether the kiln ever displayed the banner, is instrumented by randomized assignment to the environmental intervention. Banner display lowers bribe payments by 5.7 taka and compliance costs by 5.6 taka per 1,000 bricks. Relative to the control mean, these correspond to reductions of about 19.3% in bribes and 5.2% in compliance costs over a firing season. The effect on political donations is again not statistically significant. The presence of a compliance signal, independent of its duration, reduces illicit regulatory burdens.

In the binary case of the model, the estimate corresponds to $b_i^*(1) - b_i^*(0) = (\lambda\Delta - \beta\gamma)/\kappa < 0$, which holds when $\beta\gamma > \lambda\Delta$. As in the duration specification, the negative and significant effect on bribes places the setting in the region where the compliance-signaling channel outweighs the perceived-surplus channel, so that officials extract less from visibly compliant kilns rather than more.

Table 6.2: 2SLS Estimates of Banner Display on Illicit Payments per 1,000 Bricks

	<i>Bribe Payments</i>	<i>Compliance Costs</i>	<i>Political Donations</i>
Banner Displayed = 1	-5.706** (2.279)	-5.560* (3.232)	-0.839 (1.048)
Constant	40.882*** (5.473)	142.516*** (13.394)	12.852*** (1.974)
Observations	315	315	315
R-squared	0.188	0.288	0.207
Control Mean	29.58	107.6	12.81

Notes: Two-stage least squares estimates. Banner display is instrumented by randomized assignment to the environmental intervention. The binary treatment equals 1 if the kiln ever displayed the banner. All specifications include subdistrict fixed effects, kiln characteristics, and baseline illicit costs. Standard errors clustered at the environmental-intervention randomization unit are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

6.2 Robustness Checks and Heterogeneity Analysis

This section presents robustness checks and evidence on the underlying mechanism. First, we estimate the baseline specifications including total production as an additional control to assess sensitivity to scale effects. Second, we restrict the sample to kilns that did not adopt the full set of recommended technologies under the broader RCT but nevertheless received a banner after adopting the two specific practices required for eligibility.

6.2.1 Controlling for Production Scale Effects

Although Brooks et al. (2025) find no statistically significant effect of the environmental intervention on total annual brick production, bribe burdens may vary systematically with firm size and can partly offset regulatory distortions (Tamkoç, 2024). We therefore re-estimate the IV specifications including total annual production (measured in lakh bricks) as an additional control to assess sensitivity to differences in production scale across kilns.

Table 6.3 reports the results. Production scale is negatively associated with illicit costs, consistent with economies of scale in rent extraction. The effect on bribes remains negative and significant in both the duration (-0.216, from -0.294) and binary (-4.205, from -5.706) specifications, attenuated by roughly a quarter (13.6% and 14.2%) but stable in sign. The effect on compliance cost loses significance in both specifications. Since bribes are our primary measure of discretionary rent extraction, the reduction in bribes indicates that the reduction is not driven solely by changes in production capacity.

Table 6.3: IV Estimates of Banner Effect on Illicit Payments Controlling for Total Production

	<i>Bribe Payments</i>	<i>Compliance Costs</i>	<i>Political Donations</i>
<i>Panel A. Banner Duration (Weeks)</i>			
Banner Duration (Weeks)	-0.216** (0.108)	0.013 (0.173)	-0.027 (0.060)
Annual Production (Lakh)	-0.515*** (0.124)	-1.967*** (0.121)	-0.172*** (0.030)
Constant	65.566*** (1.087)	246.227*** (12.409)	21.313*** (1.279)
Observations	315	315	315
R-squared	0.280	0.538	0.274
Control mean (week)	1.58	5.76	0.69
<i>Panel B. Banner Display (Binary)</i>			
Banner Display = 1	-4.205** (1.928)	0.465 (3.342)	-0.414 (1.128)
Annual Production (Lakh)	-0.529*** (0.123)	-1.968*** (0.121)	-0.174*** (0.028)
Constant	67.092*** (0.618)	246.040*** (12.713)	21.441*** (1.485)
Observations	315	315	315
R-squared	0.281	0.538	0.274
Control Mean	29.58	107.6	12.81

Notes: 2SLS estimates. The instrument is randomized assignment to the environmental intervention. All models control for subdistrict fixed effects, kiln characteristics, and baseline illicit costs. Standard errors clustered at the intervention unit level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

6.2.2 Signaling Effects among Technology Non-Adopters

To further assess the role of signaling, we estimate the effect of banner exposure among kilns that did not continue the recommended technological practices. During banner distribution, eligibility was based on field verification of the two key practices. Sustained adoption was subsequently assessed by enumerator observation during the kiln performance assessment, following the adoption definition in Brooks et al. (2025), under which a kiln counts as an adopter only if it is observed following both improved brick stacking and single-fireman continuous feeding. We classify kilns on this observed measure and restrict the sample to those not sustaining both practices. This allows us to examine whether banner exposure reduces illicit payments even in the absence of sustained technological compliance. The relevant variation is summarized in Table A.4.

If the banner operates by altering bureaucratic perceptions of compliance, it should reduce rent extraction even among kilns that fail to maintain the practices. Table 6.4 shows that, among technology non-adopters, banner exposure significantly reduces bribe payments in both duration and binary specifications (Panel A and B, respectively). The estimated effects on political donations are also negative

and statistically significant, while compliance cost coefficients are negative but imprecisely estimated.

Because these firms do not maintain technological compliance, the observed reduction in bribes cannot be attributed to technology adoption. Instead, the findings are consistent with the interpretation that banner display reduces illicit payments by altering officials' perceived compliance status rather than through the direct effects of technology adoption.

Table 6.4: IV Estimates of Banner Effect on Illicit Payments among Technology Non-adopters

	<i>Bribe Payments</i>	<i>Compliance Costs</i>	<i>Political Donations</i>
<i>Panel A. Banner Duration (Weeks) x Technology Non-adopters</i>			
Banner Duration (Weeks)	-1.954*** (0.481)	-0.864 (1.631)	-0.463* (0.264)
Constant	114.726*** (11.314)	181.608*** (20.989)	28.144*** (5.261)
Observations	152	152	152
R-squared	0.265	0.289	0.296
Implied Weekly Cost	1.77	6.24	0.74
<i>Panel B. Banner Display x Technology Non-adopters</i>			
Banner Display = 1	-35.658*** (8.926)	-13.025 (30.851)	-8.001* (4.748)
Constant	122.114*** (11.883)	182.139*** (27.149)	29.415*** (6.007)
Observations	152	152	152
R-squared	0.221	0.268	0.245
Control Mean	33	116.6	13.79

Notes: 2SLS estimates. The instrument is randomized assignment to the environmental intervention. All models control for subdistrict fixed effects, kiln characteristics, and baseline illicit costs. Standard errors clustered at the intervention unit level in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

6.2.3 Heterogeneity by Environmental Intervention and Technology Adoption

Table A.5 presents 2SLS estimates comparing kilns at the two extremes of the distribution, those that neither received the environmental intervention nor adopted the technology (Column 1 in Table A.4) and those that received the intervention and fully adopted the technology (Column 4). Among this sample, each additional week of banner display significantly reduces bribe payments (-0.911), compliance costs (-2.066), and political donations (-0.289). Table A.6 restricts the comparison to a more focused subsample, kilns that neither received the intervention nor adopted the technology, or received the banner (Column 1 in Table A.4) and those that received the intervention, adopted the technology, and displayed the banner (Column 4, bottom cell). The results remain robust with the estimated effects of banner duration on bribes (-0.745), compliance costs (-1.799), and political donations (-0.266). The binary specifications in both subsamples yield estimates consistent with previous findings, confirming

that both the intensity and the presence of the signal are associated with lower illicit payments.

7 Conclusion

This study provides new evidence on how signaling environmental compliance shapes illicit payment demands in an informal and weakly regulated industry. We show that credible compliance signaling significantly reduces bribes and other illicit payments extracted by local officials.

The implications extend well beyond the brick kiln sector. Structural informality and weak enforcement are pervasive across industries such as garments, battery recycling, tanneries, and small-scale manufacturing in many developing countries. These sectors share common features—bureaucratic discretion, entrenched corruption, and environmental non-compliance—that sustain rent-seeking. Our findings suggest that credible, low-cost compliance signals can alter the strategic interaction between firms and regulators, reducing illicit payments even in the absence of strong institutional enforcement.

This study also highlights how the effectiveness of technological interventions depends not only on firm adoption but also on official behavior. As shown in Brooks et al. (2025), adoption of the zigzag 2.0 method improved energy efficiency and brick quality and reduced emissions, without changing total brick production. However, had officials responded by extracting higher bribes from more profitable firms, the environmental and productivity benefits would likely have been undermined. Instead, compliance signaling worked in tandem with technology adoption to reduce bribery, thereby amplifying the intervention’s impact.

That said, we cannot fully observe how bribes were distributed across different levels of bureaucracy. While our analysis focuses on interactions with local officials, it remains unclear whether and how these illicit payments are shared up the hierarchy. In settings of structural informality and weak enforcement, it may be unrealistic to expect corruption to be eliminated entirely. What our results show is that a market-driven mechanism in which producers themselves provide credible compliance signals can mitigate some of the costs of corruption and enhance firm profitability through lower fuel and illicit costs.

From a practical standpoint, more than 90% of bricks were manufactured in highly polluting fixed chimney kilns as of 2011 (World Bank, 2011), and the enormous demand for bricks to support Bangladesh’s expanding infrastructure makes shutting down all informal kilns impractical. Instead, adopting new and cost-effective technologies such as Zigzag 2.0, along with credible public signals of compliance, can reduce environmental harm while creating a “win-win” equilibrium. In such a system, regulators face lower incentives to extract bribes, firms gain from reduced compliance costs, and society benefits from lower pollution.

References

- AMIN, M. AND Y. C. SOH (2021): "Does greater regulatory burden lead to more corruption? Evidence using firm-level survey data for developing countries," *The World Bank Economic Review*, 35, 812–828.
- BANERJEE, A., S. MULLAINATHAN, AND R. HANNA (2012): "Corruption," Tech. rep., National Bureau of economic research.
- BLACKMAN, A. (2010): "Alternative Pollution Control Policies in Developing Countries," *Review of Environmental Economics and Policy*, 4, 234–253.
- BROOKS, N., D. BISWAS, S. MAITHEL, S. KUMAR, M. R. UDDIN, S. AHMED, M. MAHZAB, G. MILLER, M. RAHMAN, AND S. P. LUBY (2024): "Building blocks of change: The energy, health, and climate co-benefits of more efficient brickmaking in Bangladesh," *Energy Research & Social Science*, 117, 103738.
- BROOKS, N., D. BISWAS, S. MAITHEL, G. MILLER, A. MAHAJAN, M. R. UDDIN, S. AHMED, M. MAHZAB, M. RAHMAN, AND S. P. LUBY (2025): "Reducing emissions and air pollution from informal brick kilns: Evidence from Bangladesh," *Science*, 388, eadr7394.
- COLONNELLI, E. AND M. PREM (2022): "Corruption and firms," *The Review of Economic Studies*, 89, 695–732.
- DJANKOV, S., R. LA PORTA, F. LOPEZ-DE SILANES, AND A. SHLEIFER (2002): "The regulation of entry," *The quarterly Journal of economics*, 117, 1–37.
- DRANOVE, D. AND G. Z. JIN (2010): "Quality disclosure and certification: Theory and practice," *Journal of economic literature*, 48, 935–963.
- DUFLO, E., M. GREENSTONE, R. PANDE, AND N. RYAN (2013): "Truth-telling by third-party auditors and the response of polluting firms: Experimental evidence from India," *The Quarterly Journal of Economics*, 128, 1499–1545.
- (2018): "The value of regulatory discretion: Estimates from environmental inspections in India," *Econometrica*, 86, 2123–2160.
- HA, V. T. C., T. NGUYEN CHAU, T. T. T. PHAM, AND D. NGUYEN (2024): "Does corruption grease or sand the wheels of firm productivity? Evidence of a non-linear relationship from a firm-level analysis in Vietnam," *Journal of Economic Studies*, 51, 919–941.
- ISLAM, N. (2016): *Corruption, Its Control and Drivers of Change: The Case of Bangladesh*, Dhaka, Bangladesh: Prothoma Prokashan, first prothoma edition ed., originally published in 2014 by the Bangladesh Institute of Development Studies (BIDS), Dhaka.

- KOZŁOWSKI, S. (2021): "Climate Change, Forced Labour, and the International Labour Organization in Cambodia: Disentangling Decent Work, Economic Growth, and Environmental Damage," AP16999 Major Research Paper, Graduate School of Public and International Affairs, University of Ottawa.
- LIZZERI, A. (1999): "Information revelation and certification intermediaries," *The RAND journal of Economics*, 214–231.
- MATTSSON, M. (2025): "Information systems, service delivery, and corruption: Evidence from the bangladesh civil service," *American Economic Journal: Applied Economics*, 17, 414–445.
- MILLER, G., D. BISWAS, A. MAHAJAN, K. S. BABIARZ, N. R. BROOKS, J. BRUNNER, S. ASHRAF, J. SHANE, S. MAITHEL, S. AHMED, M. MAHZAB, M. R. UDDIN, M. RAHMAN, AND S. P. LUBY (2024): "A Business Case for Human Rights at Work? Experimental Evidence on Labor Trafficking and Child Labor at Brick Kilns in Bangladesh," *NBER Working Paper*.
- OLKEN, B. A. (2007): "Monitoring corruption: evidence from a field experiment in Indonesia," *Journal of political Economy*, 115, 200–249.
- (2009): "Corruption perceptions vs. corruption reality," *Journal of Public economics*, 93, 950–964.
- OLKEN, B. A. AND P. BARRON (2009): "The simple economics of extortion: evidence from trucking in Aceh," *Journal of Political Economy*, 117, 417–452.
- OLKEN, B. A. AND R. PANDE (2012): "Corruption in developing countries," *Annu. Rev. Econ.*, 4, 479–509.
- REINIKKA, R. AND J. SVENSSON (2004): "Local capture: evidence from a central government transfer program in Uganda," *The quarterly journal of economics*, 119, 679–705.
- (2006): "Using micro-surveys to measure and explain corruption," *World Development*, 34, 359–370.
- RODRÍGUEZ-POSE, A. AND M. DI CATALDO (2015): "Quality of government and innovative performance in the regions of Europe," *Journal of economic geography*, 15, 673–706.
- SHLEIFER, A. AND R. W. VISHNY (1993): "Corruption," *The quarterly journal of economics*, 108, 599–617.
- SPENCE, M. (1973): "Job Market Signaling," *Quarterly Journal of Economics*, 87, 355–374.
- SVENSSON, J. (2003): "Who must pay bribes and how much? Evidence from a cross section of firms," *The quarterly journal of economics*, 118, 207–230.
- TAMKOÇ, M. N. (2024): "Bribery, plant size and size dependent distortions," *Journal of Development Economics*, 171, 103361.

WORLD BANK (2011): "Introducing Energy-efficient Clean Technologies in the Brick Sector of Bangladesh," Report 60155-BD, The World Bank, Environment, Climate Change, and Water Resources Unit, South Asia Region, Washington, DC, energy Sector Management Assistance Program (ESMAP).

——— (2012): "Operationalizing Political Economy: Brick Industry in Bangladesh," South Asia Political Economy and Governance Issues Note 2, The World Bank, Washington, DC, report No. 69366.

A Appendix: Figures and Tables

Figure A.1: Siting Provisions of the 2013 Brick Manufacturing and Kiln Establishment Act

The Brick Manufacturing and Kiln Establishment (Control) Act, 2013 .pdf	
8. Prohibition on and control of establishment of brick kilns in several places.- (1) Notwithstanding anything contained in any other law for the time being in force, whether there is any clearance certificate or not, no person shall, after the enforcement of this Act, establish any brick kiln within the boundaries of the following areas, namely:-	
(a) residential, reserved or commercial areas; (b) City Corporation, Paurashava or Upazila Sadars; (c) public or privately owned forests, sanctuary, gardens or wetlands; (d) agricultural lands; (e) Ecologically Critical Area; (f) Degraded Air Shed.	
(2) After the enforcement of this Act, the Department of Environment, or any other authority under any law shall not, give permit or clearance certificate or license, by whatever name called, for establishing brick kilns within the prohibited areas.	
(3) ¹⁷*** No person shall establish brick kilns in the following distances or places, namely:-	
(a) within minimum 1 (one) kilometer distance, from the boundaries of the prohibited areas; (b) ¹⁸*** within 2 (two) kilometers distance from the boundaries of public forest; (c) in case of the establishment of brick kiln on the top or slope or the surrounding ground surface of any hill or hillock, within minimum ½ (half) kilometer distance from the foot of the said hill or hillock; (d) in case of the establishment of brick kilns in the hill districts, in any place other than the place determined by the hill districts Environment Development Committee; (e) within minimum 1 (one) kilometer distance, from any special structure, railways, educational institutions, hospitals and clinics, research institutions, or any other similar place or institution¹⁹[:]	
²⁰***]	
²¹[Provided that prior to the enactment of this Act, if the level of air pollution in the existing brick kiln remains within the prescribed standards according to the Environment Conservation Rules, 1997 and its working environment is healthy, the Government may, by notification in the official Gazette, relax the conditions for this brick kiln other than the provisions of sections 8(1)(e) and 8(3) (b), (c) and (d).]	
²²***]	
(4) Before the enforcement of this section, if any clearance certificate receiver has already established brick kilns within the boundaries of the prohibited areas or in a distance mentioned in sub-section (3), he shall, according to the provisions of this Act, transfer the said	
brick kiln in a proper place within 2 (two) years from the enforcement of this Act, otherwise his license shall be cancelled.	
Explanation: In this section,-	
(a) “residential area” means such area where at least 50(fifty) families resides; (b) “wetland” means any land which remains submerged under the water for 6 (six) months or more in a year ²³***]; (c) “Ecologically Critical Area” means any area declared under section 5 of the Environment Conservation Act; (d) “garden” means such place where there are at least 100 (one hundred) fruit or forest grown or both kind of trees in per hectare and also includes tea garden; ²⁴***] (e) “privately owned forest” means such forest which is recognized by the Forest Department as privately owned forest and whose crown cover is expanding to at least 30 (thirty) percent area of the forest, and also includes social forest or village forest²⁵[:] ²⁶[(f) “Sanctuary” means any area declared as sanctuary by notification in the Official Gazette under the power conferred by section 13 of the Wildlife (Conservation and Control) Act, 2012; (g) “Degraded Air Shed” means any area of the same air flow that exceeds the standards of air pollution set under the Environment Conservation Rules, 1997; and (h) “Public Forest” means any forest other than the forests mentioned in clause (e).]	

Figure A.2: Brick Kiln Locations

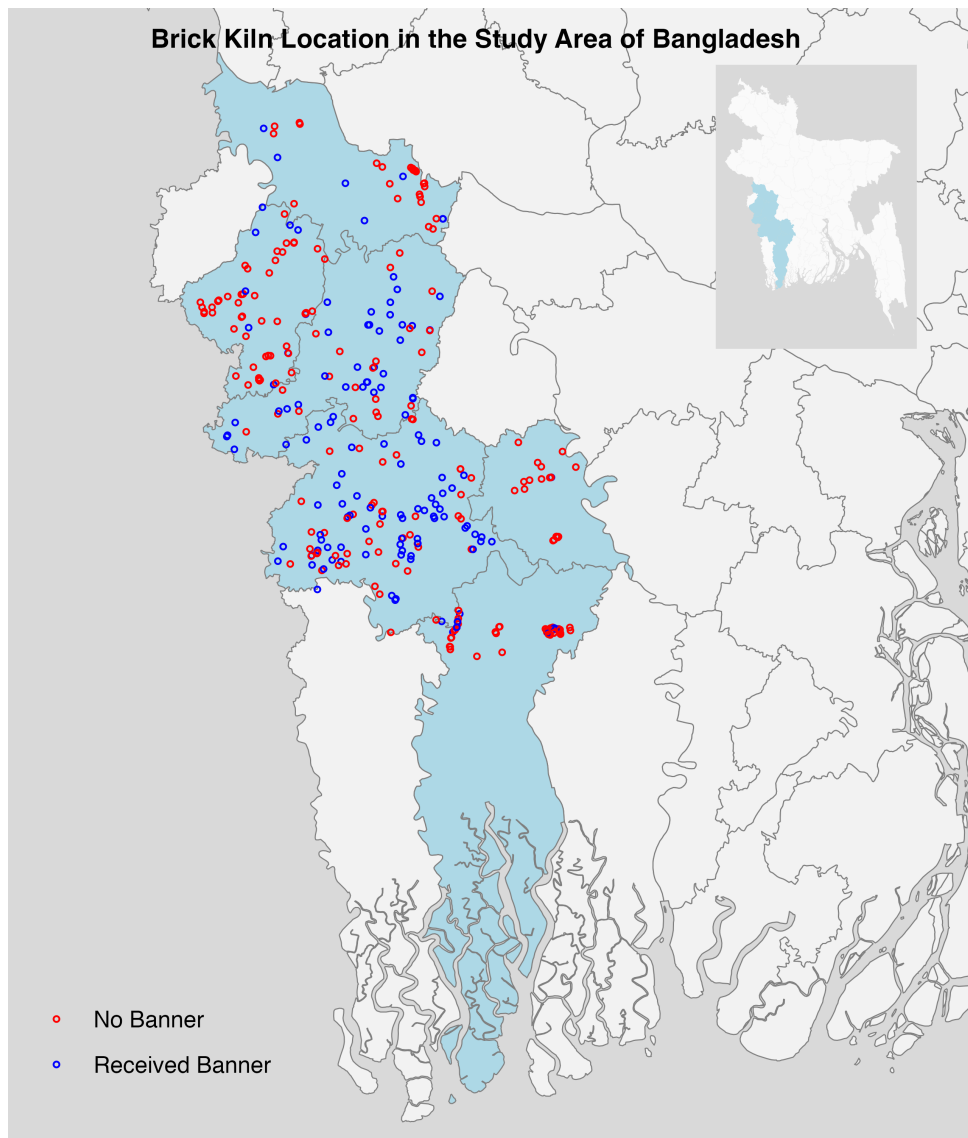


Figure A.3: Field photo during the distribution and installation of banners at participating kilns.



Note: A collection of photos taken during our intervention and also later at the scale-up stage in different districts in Bangladesh.

Figure A.4: Study Timeline

	2022				2023							
	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Oct
Brick Firing Season												
Firing End Date												
Banner Distribution												
Baseline/Endline Survey												
Kiln Charecteristics Survey												
Public Official's Interview												

Figure A.5: Study Design and Data

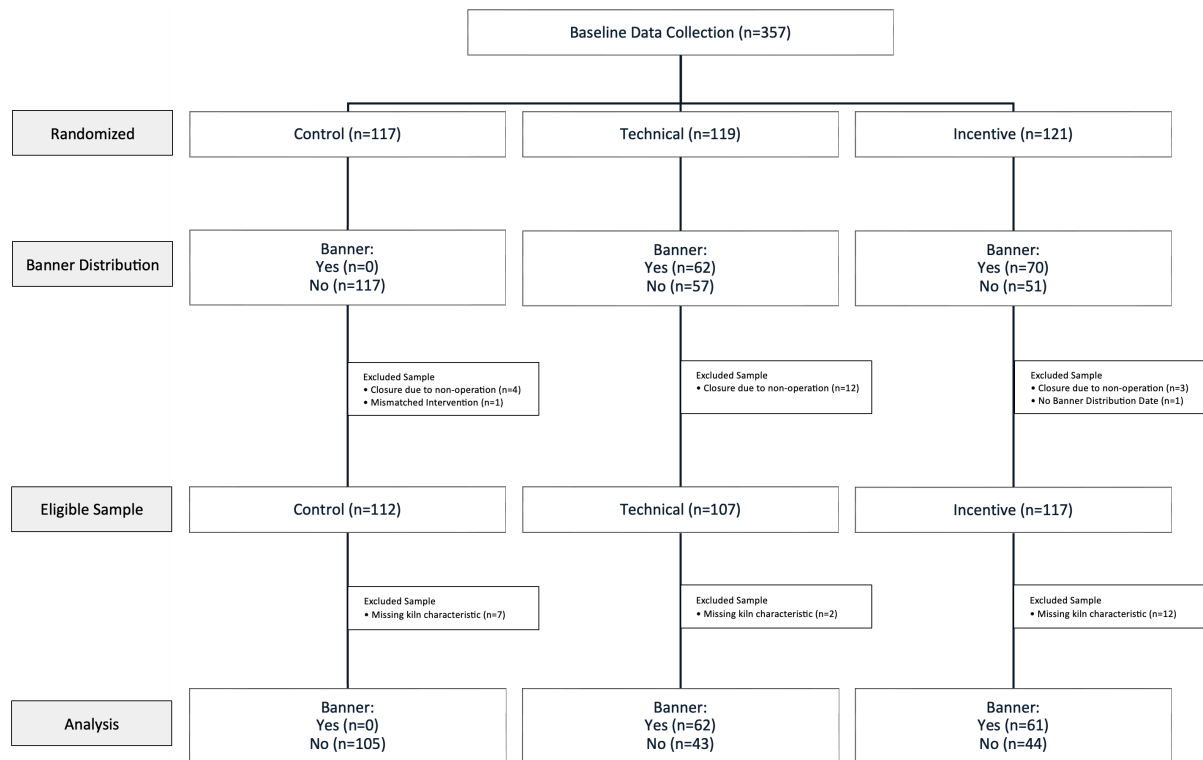


Table A.1: Interviews Conducted with Government Officials

District	District Commissioner Office	Upazila Nirbahi Office	Department of Environment	Water Development Board	Labor Department Office	Total
Jashore	1	2	2	1	1	7
Khulna	1	2	3	-	-	6
Jhenaidah	1	2	2	-	-	5
Kushtia	1	1	2	-	-	4
Chuadanga*	1	1	-	-	-	2
Total	5	8	9	1	1	24

*Chuadanga falls under the jurisdiction of the Kushtia DoE office.

Table A.2: Pre-Treatment Covariate Balance by Banner Receipt

Variable	Received Banner = 0		Received Banner = 1		Pairwise t-test
	N	Mean	N	Mean	P-value
Kiln characteristics					
Educational Institute Distance	192	1527.47 (67.15)	123	1750.41 (91.95)	0.98
Waterbody Distance	192	1860.78 (136.93)	123	2257.40 (162.31)	0.01***
Forest or Garden Distance	192	612.26 (51.67)	123	661.90 (78.35)	0.66
Residential Area Distance	192	369.87 (28.56)	123	413.12 (38.20)	0.31
Farmland Distance	192	200.34 (52.27)	123	81.33 (20.19)	0.43
Plants and Trees Coverage (%)	192	21.49 (1.24)	123	28.59 (1.58)	0.10*
Waste Management Septic Tank	192	0.01 (0.01)	123	0.01 (0.01)	0.70
Demolished Kiln	191	0.06 (0.02)	123	0.09 (0.03)	0.37
Baseline outcomes					
Bribes-1k br	192	45.69 (2.91)	123	49.93 (4.77)	0.31
Compliance-1k br	192	153.13 (5.98)	123	161.54 (10.41)	0.49
Pol. Donations-1k br	192	34.95 (2.48)	123	29.45 (2.55)	0.11
F-test of joint significance (P-value); N = 314				1.57	(0.11)
Standard errors in parentheses. District fixed effect is used for all estimations.					

Table A.3: Pre-Treatment Covariate Balance by Environmental Intervention Assignment

Variable	Intervention = 0		Intervention = 1		Pairwise t-test
	N	Mean	N	Mean	P-value
Kiln characteristics					
Educational Institute Distance	105	1609.29 (98.87)	210	1617.15 (65.65)	0.97
Waterbody Distance	105	2261.98 (200.49)	210	1892.49 (121.34)	0.02**
Forest or Garden Distance	105	666.54 (73.81)	210	614.19 (54.54)	0.55
Residential Area Distance	105	367.46 (32.22)	210	396.41 (30.40)	0.58
Farmland Distance	105	222.50 (90.47)	210	119.55 (19.83)	0.14
Plants and Trees Coverage (%)	105	24.77 (1.71)	210	24.01 (1.22)	0.33
Waste Management Septic Tank	105	0.00 (0.00)	210	0.01 (0.01)	0.33
Demolished Kiln	105	0.10 (0.03)	209	0.06 (0.02)	0.28
Baseline outcomes					
Bribes-1k br	105	45.29 (3.62)	210	48.37 (3.40)	0.57
Compliance-1k br	105	151.29 (6.11)	210	158.98 (7.59)	0.50
Pol. Donations-1k br	105	37.17 (3.70)	210	30.62 (1.98)	0.08*
F-test of joint significance (P-value); N = 314				1.36	(0.19)
Standard errors in parentheses. District fixed effects are used for all estimations.					

Table A.4: Banner Receipt and Technology Adoption in Intervention Arms.

Banner Received	Environmental Intervention & Technology Adoption				Total
	NN	NY	YN	YY	
	(1)	(2)	(3)	(4)	
No	60	45	72	15	192
Yes	0	0	20	103	123
Total	60	45	92	118	315

Note: The table categorizes the kilns in the study sample by the interaction of environmental intervention assignment and technology adoption (columns), and banner receipt (rows). The technology adoption column represents the adoption of all recommended practices by the broader intervention. Columns correspond to the four possible assignment-adoption combinations: (1) **NN**: no intervention, no technology adoption; (2) **NY**: no intervention, technology adopted; (3) **YN**: intervention assigned, no technology adoption; and (4) **YY**: intervention assigned, technology adopted. Banner distribution was conditional on the assessment of the adoption of the two specific conditions (1) proper brick stacking to improve airflow and combustion, and (2) the use of a single fireman to manage kiln firing, which enhances combustion efficiency; no kilns in the non-intervention arms (columns 1 and 2) received banners. For robustness, we compare columns (1) and (3) kilns without full technology adoption that differ only by intervention assignment, to isolate the signaling effect.

Table A.5: IV Estimates of Banner on Illicit Payments (Column (1) and (4) in Tab A.4).

	<i>Bribe Payments</i>	<i>Compliance Costs</i>	<i>Political Donations</i>
<i>Panel A. Banner Duration (Weeks)</i>			
Banner Duration (Weeks)	-0.911*** (0.195)	-2.066*** (0.182)	-0.289*** (0.045)
Constant	49.450*** (7.538)	158.630*** (4.945)	17.541*** (2.148)
Observations	178	178	178
R-squared	0.317	0.386	0.328
Weekly Mean Cost	1.686	5.716	0.733
<i>Panel B. Banner Display (Binary)</i>			
Banner Displayed = 1	-18.567*** (3.157)	-40.590*** (3.514)	-5.670*** (0.850)
Constant	54.500*** (7.207)	167.575*** (6.324)	18.878*** (2.207)
Observations	178	178	178
R-squared	0.328	0.373	0.323
Control Mean	32.16	109	13.98

Notes: 2SLS estimates. The instrument is randomized assignment to the environmental intervention. All models include sub-district fixed effects, kiln characteristics, and baseline illicit costs. Standard errors clustered at the environmental treatment unit level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A.6: IV Estimates of Banner on Illicit Payments (sample 60-103 in Tab A.4)

	<i>Bribe Payments</i>	<i>Compliance Costs</i>	<i>Political Donations</i>
<i>Panel A. Banner Duration (Weeks)</i>			
Banner Duration (Weeks)	-0.745*** (0.188)	-1.799*** (0.272)	-0.266*** (0.065)
Constant	46.759*** (5.713)	144.101*** (7.894)	14.739*** (2.854)
Observations	163	163	163
R-squared	0.331	0.404	0.344
Weekly Mean Cost	1.794	5.953	0.761
<i>Panel B. Banner Display (Binary)</i>			
Banner Displayed = 1	-14.903*** (3.201)	-34.432*** (5.579)	-5.053*** (1.327)
Constant	50.388*** (5.699)	150.376*** (10.249)	15.692*** (3.231)
Observations	163	163	163
R-squared	0.340	0.391	0.338
Control Mean	34.23	113.6	14.51

Notes: 2SLS estimates. The instrument is randomized assignment to the environmental intervention. All models include sub-district fixed effects, kiln characteristics, and baseline illicit costs. Standard errors clustered at the environmental treatment unit level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.